

Università
degli Studi
di Padova

COLLECTION
OF

Theorems and Proofs for Machine Learning

Created By

Gabriele Bellavia

University of Padua
Academic year 2025–2026

October 12, 2026

About these notes

These notes are part of my `PoD-collection-of-notes` repository and were written during my MSc in Physics of Data at the University of Padua between 2024 and 2026.

They are intended as freely accessible study material and may contain errors, inaccuracies, or outdated information. Most of the material was written while attending the courses and was not systematically reviewed afterwards. Course contents and syllabi may also change over time.

License

Except where otherwise noted, this work is licensed under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License (CC BY-NC-SA 4.0).

You are free to share and adapt this material for non-commercial purposes, provided that appropriate credit is given, a reference to the original source is included, and any derivative work is distributed under the same license.

Source files

The complete source files used to compile these notes, including figures and other resources, are available in the corresponding folder of the repository:

`https://github.com/gabriele-bellavia/PoD-collection-of-notes`

Citation

If you use these notes, redistribute them, or use them as a starting point for your own material, please cite the original repository and, where applicable, the authors of any external material referenced in these notes.

Errors and corrections

If you find an error, typo, or unclear explanation, please feel free to open an issue on the GitHub repository or contact me at:

`gabriele.bellavia@studenti.unipd.it`

Physics of Data MSc
University of Padua

Contents

1 Theorems and Definitions **1**
1.1 Learning Framework Notation 1
1.2 PAC Learning Theorem 1
1.3 Representativeness Lemma 2
1.4 Agnostic PAC Learnability of Finite Hypothesis Classes Theorem 3
1.5 The fundamental theorem of PAC learning 5
1.6 Structural Risk Minimization 7
1.7 Perceptron Convergence 8
1.8 VC-dimension of Halfspaces 9
1.9 Optimality of the Least Squares Algorithm 10
1.10 Distance Theorem for Halfspaces 11
1.11 Equivalence of the Convex SVM Problem 11
1.12 Representer Theorem 12
1.13 Polynomial Kernel Derivation 13
1.14 RBF Kernel Derivation 13
1.15 Weak Learnability Property of Decision Stumps 14
1.16 VC-dimension of Boosted Solutions 14
1.17 SGD Bound for RLM 15
1.18 Derivation of Backpropagation 17
1.19 Optimal Inverse Lemma for PCA 19
1.20 Correctness of K-means Clustering 19

Chapter 1

Theorems and Definitions

1.1 Learning Framework Notation

1.2 PAC Learning Theorem

Definition 1.2.1 (PAC Learnability)**.** A hypothesis class $\boldsymbol{\mathcal{H}}$ is **PAC learnable** if there exists a function $m_{\boldsymbol{\mathcal{H}}} : (0,1)^2 \to \mathbb{N}$ and a learning algorithm with properties: for every $\varepsilon, \delta \in (0,1)$, for every distribution $\boldsymbol{\mathcal{D}}$ over $\boldsymbol{\mathcal{X}}$, and for every labeling function $f : \boldsymbol{\mathcal{X}} \to \{0,1\}$, if the realizable assumption holds with respect to $\boldsymbol{\mathcal{H}}$, $\boldsymbol{\mathcal{D}}$, f, then when running the learning algorithm on $m \geq m_{\boldsymbol{\mathcal{H}}}(\varepsilon, \delta)$ iid examples generated by $\boldsymbol{\mathcal{D}}$ and labeled by f, the algorithm returns a hypothesis h such that, with probability of at least $1 - \delta$ over the choice of the examples, $L_{\boldsymbol{\mathcal{D}},f}(h) \leq \varepsilon$.

Theorem 1.2.2 (PAC Learning Theorem)**.** *Let $\boldsymbol{\mathcal{H}}$ be a finite hypothesis class. Let $\delta, \varepsilon \in (0,1)$ and $m \in \mathbb{N}$ such that*

$$m \geq \frac{1}{\varepsilon} \log\left(\frac{|\boldsymbol{\mathcal{H}}|}{\delta}\right) \tag{1.1}$$

Then, for any labeling function f and for any iid distribution $\boldsymbol{\mathcal{D}}$ for which realizability assumption holds, we have that any $h_{\boldsymbol{\mathcal{S}}}$ computed with ERM over training set $\boldsymbol{\mathcal{S}}$ of size m respects

$$L_{\mathcal{D},f}(h_{\boldsymbol{\mathcal{S}}}) \leq \varepsilon \tag{1.2}$$

with probability greater than $1 - \delta$.

Proof. Let us define the set of bad hypotheses

$$\boldsymbol{\mathcal{H}}_B = \{h \in \boldsymbol{\mathcal{H}} : L_{\mathcal{D},f}(h) > \varepsilon\}$$

and the set of training sets in which a bad hypothesis works

$$M = \{\boldsymbol{\mathcal{S}} \in X^m : \exists h \in \boldsymbol{\mathcal{H}}_B, L_{\boldsymbol{\mathcal{S}}}(h) = 0\}$$

We know that, in order for ERM to fail, we must be in M (remember, ERM selects one of the hypotheses with zero training loss):

$$p(L_{\boldsymbol{\mathcal{D}},f}(h_{\boldsymbol{\mathcal{S}}}) > \varepsilon) \leq \boldsymbol{\mathcal{D}}^m(M)$$

where we used the iid hypothesis over $\boldsymbol{\mathcal{D}}$. The first step is rewriting set M as

$$M = \bigcup_{h \in \boldsymbol{\mathcal{H}}_B} \{\boldsymbol{\mathcal{S}} \in X^m : L_{\boldsymbol{\mathcal{S}}}(h) = 0\}$$

and applying the union bound $\boldsymbol{\mathcal{D}}^m(A \cup B) \leq \boldsymbol{\mathcal{D}}^m(A) + \boldsymbol{\mathcal{D}}^m(B)$, we have

$$p(L_{\boldsymbol{\mathcal{D}},f}(h_{\boldsymbol{\mathcal{S}}}) > \varepsilon) \leq \boldsymbol{\mathcal{D}}^m(M) \leq \sum_{h \in \boldsymbol{\mathcal{H}}_B} \boldsymbol{\mathcal{D}}^m(\{\boldsymbol{\mathcal{S}} : L_{\boldsymbol{\mathcal{S}}}(h) = 0\})$$

Therefore, the second step is, saying that a hypothesis has zero training loss is equivalent to saying it correctly classifies all point in the training set:

$$\boldsymbol{\mathcal{D}}^m(\{\boldsymbol{\mathcal{S}} : L_{\boldsymbol{\mathcal{S}}}(h) = 0\}) = \prod_{i=1}^{m} \boldsymbol{\mathcal{D}}(\{x_i : h(x_i) = f(x_i)\})$$

By definition of a bad hypothesis, we have

$$\boldsymbol{\mathcal{D}}(\{x : h(x) = f(x)\}) < 1 - \varepsilon$$

The third step is that as the sample is iid, we know that

$$\boldsymbol{\mathcal{D}}^m(\{\boldsymbol{\mathcal{S}} : L_{\boldsymbol{\mathcal{S}}}(h) = 0\}) < (1 - \varepsilon)^m$$

We can use inequality $(1 - x) \leq e^{-x}$ to get

$$\boldsymbol{\mathcal{D}}^m(\{\boldsymbol{\mathcal{S}} : L_{\boldsymbol{\mathcal{S}}}(h) = 0\}) < e^{-\varepsilon m}$$

Since we need to sum over $\boldsymbol{\mathcal{H}}_B$, we get

$$\boldsymbol{\mathcal{D}}^m(M) < |\boldsymbol{\mathcal{H}}_B| e^{-\varepsilon m} < |\boldsymbol{\mathcal{H}}| e^{-\varepsilon m}$$

where in the last inequality, we used the realizability assumption, for which there is at least one good hypothesis.

Putting it all together, we just proved that

$$p(L_{\boldsymbol{\mathcal{D}},f}(h_{\boldsymbol{\mathcal{S}}}) > \varepsilon) \leq \boldsymbol{\mathcal{D}}^m(M) < |\boldsymbol{\mathcal{H}}| e^{-\varepsilon m}$$

We can then guarantee condition $p(L_{\boldsymbol{\mathcal{D}},f}(h_{\boldsymbol{\mathcal{S}}}) > \varepsilon) < \delta$ if

$$\delta > |\boldsymbol{\mathcal{H}}| e^{-\varepsilon m}$$

This is equivalent to the theorem statement Equation 1.1. □

1.3 Representativeness Lemma

Definition 1.3.1 (ε-representativeness)**.** A training set $\boldsymbol{\mathcal{S}}$ is called ε**-representative** with respect to a domain $\boldsymbol{\mathcal{Z}}$, a hypothesis class $\boldsymbol{\mathcal{H}}$, a loss function ℓ and a data generating distribution $\boldsymbol{\mathcal{D}}$ if

$$|L_{\boldsymbol{\mathcal{D}}}(h) - L_{\boldsymbol{\mathcal{S}}}(h)| \leq \varepsilon, \qquad \forall h \in \boldsymbol{\mathcal{H}} \tag{1.3}$$

Lemma 1.3.2 (Representativeness Lemma)**.** *Assume that a training set $\boldsymbol{\mathcal{S}}$ is ε-representative with respect to a domain $\boldsymbol{\mathcal{Z}}$, a hypothesis class $\boldsymbol{\mathcal{H}}$, a loss function ℓ and a data-generating distribution $\boldsymbol{\mathcal{D}}$. Then, any ERM rule that outputs $ERM_{\boldsymbol{\mathcal{H}}}(\boldsymbol{\mathcal{S}})$, namely any $h_{\boldsymbol{\mathcal{S}}} \in \arg\min_{h \in \boldsymbol{\mathcal{H}}} L_{\boldsymbol{\mathcal{S}}}(h)$, satisfies*

$$L_{\boldsymbol{\mathcal{D}}}(h_{\boldsymbol{\mathcal{S}}}) \leq \min_{h \in \boldsymbol{\mathcal{H}}} L_{\boldsymbol{\mathcal{D}}}(h) + 2\varepsilon \tag{1.4}$$

that is the definition of PAC learnability for the hypothesis class $\boldsymbol{\mathcal{H}}$.

Proof. Let us take the best hypothesis h^* and the ERM hypothesis $h_{\boldsymbol{\mathcal{S}}}$. Due to ε-representativeness, we know that

$$L_{\boldsymbol{\mathcal{S}}}(h^*) - \varepsilon \leq L_{\boldsymbol{\mathcal{D}}}(h^*) \leq L_{\boldsymbol{\mathcal{S}}}(h^*) + \varepsilon$$

The same holds for $h_{\boldsymbol{\mathcal{S}}}$

$$L_{\boldsymbol{\mathcal{S}}}(h_{\boldsymbol{\mathcal{S}}}) - \varepsilon \leq L_{\boldsymbol{\mathcal{D}}}(h_{\boldsymbol{\mathcal{S}}}) \leq L_{\boldsymbol{\mathcal{S}}}(h_{\boldsymbol{\mathcal{S}}}) + \varepsilon$$

First step: due to the ERM rule, we know that

$$L_{\boldsymbol{\mathcal{S}}}(h_{\boldsymbol{\mathcal{S}}}) \leq L_{\boldsymbol{\mathcal{S}}}(h^*)$$

Combining this with the consequences of ε-representativeness,

$$L_{\boldsymbol{\mathcal{S}}}(h_{\boldsymbol{\mathcal{S}}}) - \varepsilon \leq L_{\boldsymbol{\mathcal{D}}}(h_{\boldsymbol{\mathcal{S}}}) \leq L_{\boldsymbol{\mathcal{S}}}(h_{\boldsymbol{\mathcal{S}}}) + \varepsilon$$

We get

$$L_{\boldsymbol{\mathcal{D}}}(h_{\boldsymbol{\mathcal{S}}}) - \varepsilon \leq L_{\boldsymbol{\mathcal{S}}}(h^*) \tag{1.5}$$

Second step: due to the consequences of ε-representativeness,

$$L_{\boldsymbol{\mathcal{S}}}(h^*) - \varepsilon \leq L_{\boldsymbol{\mathcal{D}}}(h^*) \leq L_{\boldsymbol{\mathcal{S}}}(h^*) + \varepsilon$$

Combining with Equation 1.5, we have

$$L_{\boldsymbol{\mathcal{D}}}(h_{\boldsymbol{\mathcal{S}}}) - \varepsilon \leq L_{\boldsymbol{\mathcal{D}}}(h^*) + \varepsilon$$

From which we have

$$L_{\boldsymbol{\mathcal{D}}}(h_{\boldsymbol{\mathcal{S}}}) \leq L_{\boldsymbol{\mathcal{D}}}(h^*) + 2\varepsilon = \min_{h \in \boldsymbol{\mathcal{H}}} L_{\boldsymbol{\mathcal{D}}}(h) + 2\varepsilon.$$

□

1.4 Agnostic PAC Learnability of Finite Hypothesis Classes Theorem

Definition 1.4.1 (Agnostic PAC Learnability)**.** A hypothesis class $\boldsymbol{\mathcal{H}}$ is **agnostic PAC learnable** with respect to a set $\boldsymbol{\mathcal{Z}}$ and a loss function $\ell : \boldsymbol{\mathcal{H}} \times \boldsymbol{\mathcal{Z}} \to \mathbb{R}^+$ if there exist a function $m_{\boldsymbol{\mathcal{H}}} : (0,1)^2 \to \mathbb{N}$ and a learning algorithm such that for every $\varepsilon, \delta \in (0,1)$, for every distribution $\boldsymbol{\mathcal{D}}$ over $\boldsymbol{\mathcal{Z}} \equiv \boldsymbol{\mathcal{X}} \times \boldsymbol{\mathcal{Y}}$ when running the learning algorithm on $m \geq m_{\boldsymbol{\mathcal{H}}}(\varepsilon, \delta)$ iid examples generated by $\boldsymbol{\mathcal{D}}$, the algorithm returns a hypothesis $h \in \boldsymbol{\mathcal{H}}$ such that

$$L_{\boldsymbol{\mathcal{D}}}(h) \leq \min_{h' \in \boldsymbol{\mathcal{H}}} L_{\boldsymbol{\mathcal{D}}}(h') + \varepsilon$$

where $L_{\boldsymbol{\mathcal{D}}}(h) = \mathbb{E}_{z \sim \boldsymbol{\mathcal{D}}}[\ell(h, z)]$ with a probability of at least $1 - \delta$ over the choice of the m training examples.

Definition 1.4.2 (Uniform Convergence)**.** A hypothesis class $\boldsymbol{\mathcal{H}}$ has the **uniform convergence** property with respect to a domain $\boldsymbol{\mathcal{Z}}$ and a loss function ℓ if there exists a function $m^{\text{u.c.}}_{\boldsymbol{\mathcal{H}}} : (0,1)^2 \to \mathbb{N}$ such that $\forall \varepsilon, \delta \in (0,1)$ and $\forall \boldsymbol{\mathcal{D}}$ over $\boldsymbol{\mathcal{Z}} \equiv \boldsymbol{\mathcal{X}} \times \boldsymbol{\mathcal{Y}}$, if $\boldsymbol{\mathcal{S}}$ is a sample of $m \geq m^{\text{u.c.}}_{\boldsymbol{\mathcal{H}}}(\varepsilon, \delta)$ examples drawn iid according to $\boldsymbol{\mathcal{D}}$ then $\boldsymbol{\mathcal{S}}$ is ε-representative with probability higher than $1 - \delta$.

Theorem 1.4.3 (Uniform Convergence Theorem)**.** *If a hypothesis class $\boldsymbol{\mathcal{H}}$ has the uniform convergence property with a function $m_{\boldsymbol{\mathcal{H}}}^{u.c.}$, then*

1. *The class $\boldsymbol{\mathcal{H}}$ is agnostically PAC learnable with sample complexity*

$$m_{\boldsymbol{\mathcal{H}}}(2\varepsilon, \delta) \leq m_{\boldsymbol{\mathcal{H}}}^{u.c.}(\varepsilon, \delta) \tag{1.6}$$

2. *The ERM paradigm is a valid PAC learning algorithm for $\boldsymbol{\mathcal{H}}$*

Theorem 1.4.4 (Agnostic PAC Learnability of Finite Hypothesis Classes Theorem)**.** *Let $\boldsymbol{\mathcal{H}}$ be a finite hypothesis class, let $\boldsymbol{\mathcal{Z}}$ be a domain and let $\ell : \boldsymbol{\mathcal{H}} \times \boldsymbol{\mathcal{Z}} \to [0, 1]$ be a loss function.*

1. *$\boldsymbol{\mathcal{H}}$ enjoys the uniform convergence property with sample complexity*

$$m_{\boldsymbol{\mathcal{H}}}^{u.c.}(\varepsilon, \delta) \leq \left\lceil \frac{\log\left(\frac{2|\boldsymbol{\mathcal{H}}|}{\delta}\right)}{2\varepsilon^2} \right\rceil \tag{1.7}$$

2. *$\boldsymbol{\mathcal{H}}$ is agnostic PAC learnable using the ERM algorithm with sample complexity*

$$m_{\boldsymbol{\mathcal{H}}}(\varepsilon, \delta) \leq m_{\boldsymbol{\mathcal{H}}}^{u.c.}(\varepsilon, \delta) \leq \left\lceil \frac{2\log\left(\frac{2|\boldsymbol{\mathcal{H}}|}{\delta}\right)}{\varepsilon^2} \right\rceil \tag{1.8}$$

Proof. To prove uniform convergence, it suffices to show that

$$\boldsymbol{\mathcal{D}}^m(\{\boldsymbol{\mathcal{S}} : \exists h \in \boldsymbol{\mathcal{H}} : |L_{\boldsymbol{\mathcal{S}}}(h) - L_{\boldsymbol{\mathcal{D}}}(h)| > \varepsilon\}) \leq \delta$$

which can be rewritten as the union over all hypotheses

$$\boldsymbol{\mathcal{D}}^m\left(\bigcup_{h \in \boldsymbol{\mathcal{H}}} \{\boldsymbol{\mathcal{S}} : |L_{\boldsymbol{\mathcal{S}}}(h) - L_{\boldsymbol{\mathcal{D}}}(h)| > \varepsilon\}\right) \leq \delta$$

We can apply the union bound:

$$\boldsymbol{\mathcal{D}}^m(\{\boldsymbol{\mathcal{S}} : \exists h \in \boldsymbol{\mathcal{H}} : |L_{\boldsymbol{\mathcal{S}}}(h) - L_{\boldsymbol{\mathcal{D}}}(h)| > \varepsilon\}) \leq \sum_{h \in \boldsymbol{\mathcal{H}}} \boldsymbol{\mathcal{D}}^m(\{\boldsymbol{\mathcal{S}} : |L_{\boldsymbol{\mathcal{S}}}(h) - L_{\boldsymbol{\mathcal{D}}}(h)| > \varepsilon\}) \tag{1.9}$$

We now use the Hoeffding's inequality: let $\theta_1, \ldots, \theta_m$ be a sequence of iid random variables and assume that for all i, $\mathbb{E}[\theta_i] = \mu$ and $\mathbb{P}[a \leq \theta_i \leq b] = 1$. Then, for any $\varepsilon > 0$

$$\mathbb{P}\left[\left|\frac{1}{m}\sum_{i=1}^{m}\theta_i - \mu\right| > \varepsilon\right] \leq 2\exp\left(-2m\varepsilon^2/(b-a)^2\right) \tag{1.10}$$

We can consider Hoeffding's inequality, with $\theta_i = \ell(h, z_i)$. We have $\mu = L_{\boldsymbol{\mathcal{D}}}(h)$. Let us assume that $\ell(h, z) \in [0, 1]$, so $[a, b] = [0, 1]$. Therefore, applying Equation 1.10

$$\mathbb{P}\left[\left|\frac{1}{m}\sum_{i=1}^{m}\ell(h, z_i) - L_{\boldsymbol{\mathcal{D}}}(h)\right| > \varepsilon\right] \leq 2\exp\left(-2m\varepsilon^2\right)$$

Then, we take the constant term out of the sum

$$\mathbb{P}\left[\left|\left(\frac{1}{m}\sum_{i=1}^{m}\ell(h,z_i)\right)-L_{\boldsymbol{\mathcal{D}}}(h)\right|>\varepsilon\right]\leq 2\exp\left(-2m\varepsilon^2\right)$$

We can see that the sum is simply the empirical risk:

$$\boldsymbol{\mathcal{D}}^m(\{\boldsymbol{\mathcal{S}}:|L_{\boldsymbol{\mathcal{S}}}(h)-L_{\boldsymbol{\mathcal{D}}}(h)|>\varepsilon\})\leq 2\exp\left(-2m\varepsilon^2\right)$$

Substituting in Equation 1.9, we get

$$\boldsymbol{\mathcal{D}}^m(\{\boldsymbol{\mathcal{S}}:\exists h\in\boldsymbol{\mathcal{H}}:|L_{\boldsymbol{\mathcal{S}}}(h)-L_{\boldsymbol{\mathcal{D}}}(h)|>\varepsilon\})\leq 2|\boldsymbol{\mathcal{H}}|\exp\left(-2m\varepsilon^2\right)$$

Now we need to impose to this bound an unlucky training set probability δ

$$\boldsymbol{\mathcal{D}}^m(\{\boldsymbol{\mathcal{S}}:\exists h\in\boldsymbol{\mathcal{H}}:|L_{\boldsymbol{\mathcal{S}}}(h)-L_{\boldsymbol{\mathcal{D}}}(h)|>\varepsilon\})\leq 2|\boldsymbol{\mathcal{H}}|\exp\left(-2m\varepsilon^2\right)\leq\delta$$

Isolating the training set size, we get

$$\exp\left(-2m\varepsilon^2\right)\leq\frac{\delta}{2|\boldsymbol{\mathcal{H}}|}$$

Taking the logarithm and changing the sign, we get

$$2m\varepsilon^2\geq\log\left(\frac{2|\boldsymbol{\mathcal{H}}|}{\delta}\right)$$

which is equivalent to the $m_{\boldsymbol{\mathcal{H}}}^{\text{u.c.}}$ in the theorem. Therefore, we proved that we have uniform convergence with

$$m_{\boldsymbol{\mathcal{H}}}^{\text{u.c.}}(\varepsilon,\delta)\leq\left\lceil\frac{\log\left(\frac{2|\boldsymbol{\mathcal{H}}|}{\delta}\right)}{2\varepsilon^2}\right\rceil$$

Since we already proved agnostic PAC learnability via uniform convergence, we know that $m_{\boldsymbol{\mathcal{H}}}(2\varepsilon,\delta)\leq m_{\boldsymbol{\mathcal{H}}}^{\text{u.c.}}(\varepsilon,\delta)$, so we simply get the second half of the theorem

$$m_{\boldsymbol{\mathcal{H}}}(\varepsilon,\delta)\leq m_{\boldsymbol{\mathcal{H}}}^{\text{u.c.}}\left(\frac{\varepsilon}{2},\delta\right)\leq\left\lceil\frac{2\log\left(\frac{2|\boldsymbol{\mathcal{H}}|}{\delta}\right)}{\varepsilon^2}\right\rceil$$

□

1.5 The fundamental theorem of PAC learning

Theorem 1.5.1 (No Free Lunch Theorem)**.** *Let A be any learning algorithm for the binary classification task with respect to the 0-1 loss over a domain $\boldsymbol{\mathcal{X}}$. Let m be any number smaller than $|\boldsymbol{\mathcal{X}}|/2$, representing a training set size. Then, there exists a distribution $\boldsymbol{\mathcal{D}}$ over $\boldsymbol{\mathcal{X}}\times\{0,1\}$ such that*

1. *There exists a function $f:\boldsymbol{\mathcal{X}}\to\{0,1\}$ with $L_{\boldsymbol{\mathcal{D}}}(f)=0$ (existence of a perfect solution)*
2. *we have that $L_{\boldsymbol{\mathcal{D}}}(A(\boldsymbol{\mathcal{S}}))\geq\frac{1}{8}$ with probability of at least $\frac{1}{7}$ over the choice of $\boldsymbol{\mathcal{S}}\sim\boldsymbol{\mathcal{D}}^m$.*

Corollary 1.5.2. *Let $\mathcal{X}$ be an infinite domain set and let $\mathcal{H}$ be the set of all functions from $\mathcal{X}$ to $\{0,1\}$. Then $\mathcal{H}$ is not PAC learnable.*

Definition 1.5.3 (Restriction of $\mathcal{H}$ to $\mathcal{C}$)**.** Let $\mathcal{H}$ be a class of functions from $\mathcal{X}$ to $\{0,1\}$ and let $\mathcal{C} = \{c_1, \ldots, c_m\} \subset \mathcal{X}$. The **restriction of $\mathcal{H}$ to $\mathcal{C}$** is the set of functions from $\mathcal{C}$ to $\{0,1\}$ that can be derived from $\mathcal{H}$. That is

$$\mathcal{H}_{\mathcal{C}} = \{(h(c_1), \ldots, h(c_m)) : h \in \mathcal{H}\}$$

where we represent each function from $\mathcal{C}$ to $\{0,1\}$ as a vector in $\{0,1\}^{|\mathcal{C}|}$

Definition 1.5.4 (Shattering)**.** A hypothesis class $\mathcal{H}$ **shatters** a finite set $\mathcal{C} \subset \mathcal{X}$ if the restriction of $\mathcal{H}$ to $\mathcal{C}$ is the set of all functions from $\mathcal{C}$ to $\{0,1\}$. That is $|\mathcal{H}_{\mathcal{C}}| = 2^{|\mathcal{C}|}$.

Corollary 1.5.5. *Let $\mathcal{H}$ be a hypothesis class of functions from $\mathcal{X}$ to $\{0,1\}$. Let m be a training set size. Assume that there exists a set $\mathcal{C} \subset \mathcal{X}$ of size $2m$ that is shattered by $\mathcal{H}$. Then, for any learning algorithm A there exists a distribution $\mathcal{D}$ over $\mathcal{X} \times \{0,1\}$ and a predictor $h \in \mathcal{H}$ such that $L_{\mathcal{D}}(h) = 0$ but $L_{\mathcal{D}}(A(\mathcal{S})) \geq \frac{1}{8}$ with probability of at least $\frac{1}{7}$ over the choice of $\mathcal{S} \sim \mathcal{D}^m$.*

Definition 1.5.6 (VC-dimension)**.** The Vapnik-Chervonenkis (VC) dimension of a hypothesis class $\mathcal{H}$, denoted VCdim($\mathcal{H}$), is the maximal size of a set $\mathcal{C} \subset \mathcal{X}$ that can be shattered by $\mathcal{H}$. If $\mathcal{H}$ can shatter sets of arbitrarily large size, we say that $\mathcal{H}$ has infinite VC-dimension.

Theorem 1.5.7 (Fundamental Theorem of PAC learning)**.** *Let $\mathcal{H}$ be a hypothesis class of functions from a domain $\mathcal{X}$ to $\{0,1\}$ and let the loss function be the 0-1 loss. Then, the following are equivalent:*

1. *$\mathcal{H}$ has the uniform convergence property.*
2. *Any ERM rule is a successful agnostic PAC learner for $\mathcal{H}$.*
3. *$\mathcal{H}$ is agnostic PAC learnable.*
4. *$\mathcal{H}$ is PAC learnable (under realizability condition).*
5. *Any ERM rule is a successful PAC learner for $\mathcal{H}$.*
6. *$\mathcal{H}$ has a finite VC-dimension.*

Proof. We can easily prove that $1 \Rightarrow 2 \Rightarrow 3$ from our results on uniform convergence using Theorem 1.4.3.

The implication $3 \Rightarrow 4$ is trivial, as agnostic PAC and PAC are the same if we add the realizability condition, and so is $2 \Rightarrow 5$. Thus, up to now we have $1 \Rightarrow 2 \Rightarrow 3 \Rightarrow 4$ and $1 \Rightarrow 2 \Rightarrow 5$.

To prove $4 \Rightarrow 5$ and $5 \Rightarrow 6$ we use the Corollary 1.5.2 of the No Lunch Theorem. By contradiction, if $\mathcal{H}$ has an infinite VC-dimension, it is not PAC learnable, as the theorem applies to any training set size.

The final step $6 \Rightarrow 1$ is very complex and we do not prove it.

□

1.6 Structural Risk Minimization

Definition 1.6.1 (Non-uniform learnability)**.** A hypothesis class $\boldsymbol{\mathcal{H}}$ is **non-uniformly learnable** if there exists a learning algorithm A and a function $m^{\text{n.u.}}_{\boldsymbol{\mathcal{H}}} : (0,1)^2 \to \mathbb{N}$ such that, for every $\delta, \varepsilon \in (0,1)$ and for every $h \in \boldsymbol{\mathcal{H}}$, if $m \geq m^{\text{n.u.}}_{\boldsymbol{\mathcal{H}}}(\varepsilon, \delta, h)$, with probability at least $1-\delta$ over the choice of $\boldsymbol{\mathcal{S}} \sim \boldsymbol{\mathcal{D}}^m$ for every distribution $\boldsymbol{\mathcal{D}}$, we have

$$L_{\boldsymbol{\mathcal{D}}}(A(\boldsymbol{\mathcal{S}})) \leq L_{\boldsymbol{\mathcal{D}}}(h) + \varepsilon \tag{1.11}$$

Theorem 1.6.2. *If $\boldsymbol{\mathcal{H}}$ is a hypothesis class that can be written as union $\boldsymbol{\mathcal{H}} = \bigcup_{n\in\mathbb{N}} \boldsymbol{\mathcal{H}}_n$, and each $\boldsymbol{\mathcal{H}}_n$ has uniform convergence, $\boldsymbol{\mathcal{H}}$ is non-uniformly learnable.*

Definition 1.6.3 (Upper bound gap)**.** The **upper bound gap** for a fixed training set of size m and subclass $\boldsymbol{\mathcal{H}}_n$, with uniform convergence function $m^{\text{u.c.}}_{\boldsymbol{\mathcal{H}}_n}(\varepsilon, \delta)$, is

$$\varepsilon_n(m, \delta) = \min\left\{\varepsilon \in (0,1) : m^{\text{u.c.}}_{\boldsymbol{\mathcal{H}}_n}(\varepsilon, \delta) \leq m\right\} \tag{1.12}$$

Theorem 1.6.4 (Structural Risk Bound)**.** *Let $w : \mathbb{N} \to [0,1]$ be a weight function such that $\sum_{n=1}^{\infty} w(n) \leq 1$. If $\boldsymbol{\mathcal{H}}$ is a hypothesis class that can be written as union $\boldsymbol{\mathcal{H}} = \bigcup_{n\in\mathbb{N}} \boldsymbol{\mathcal{H}}_n$, and each $\boldsymbol{\mathcal{H}}_n$ has uniform convergence with function $m^{u.c.}_{\boldsymbol{\mathcal{H}}_n}(\varepsilon, \delta)$. For every $\delta \in (0,1)$ and distribution $\boldsymbol{\mathcal{D}}$, we have the following bound for every $n \in \mathbb{N}$ and $h \in \boldsymbol{\mathcal{H}}_n$:*

$$|L_{\boldsymbol{\mathcal{D}}}(h) - L_{\boldsymbol{\mathcal{S}}}(h)| \leq \varepsilon_n(m, w(n)\delta) \tag{1.13}$$

We also have that, for any $\delta \in (0,1)$ and distribution $\boldsymbol{\mathcal{D}}$, the following bound holds with probability of at least $1-\delta$:

$$L_{\boldsymbol{\mathcal{D}}}(h) \leq L_{\boldsymbol{\mathcal{S}}}(h) + \min_{n:\, h\in\boldsymbol{\mathcal{H}}_n} \varepsilon_n(m, w(n)\delta) \qquad \forall h \in \boldsymbol{\mathcal{H}} \tag{1.14}$$

Proof. It is easy to see that the first bound in Equation 1.13 is a direct application of Equation 1.12. We then look at the second bound, and consider that proving that the accuracy is bounded by the minimum ε_n is equivalent to proving that it is bounded at the same time by the ε_n for every subclass:

$$\mathbb{P}\left[L_{\boldsymbol{\mathcal{D}}}(h) > L_{\boldsymbol{\mathcal{S}}}(h) + \min_{n:\, h\in\boldsymbol{\mathcal{H}}_n} \varepsilon_n(m, w(n)\delta)\right] = \mathbb{P}\left[\bigcup_{n:\, h\in\boldsymbol{\mathcal{H}}_n} \{L_{\boldsymbol{\mathcal{D}}}(h) > L_{\boldsymbol{\mathcal{S}}}(h) + \varepsilon_n(m, w(n)\delta)\}\right] \tag{1.15}$$

We can then apply the union bound to the right side of the equation:

$$\mathbb{P}\left[\bigcup_{n:\, h\in\boldsymbol{\mathcal{H}}_n} \{L_{\boldsymbol{\mathcal{D}}}(h) > L_{\boldsymbol{\mathcal{S}}}(h) + \varepsilon_n(m, w(n)\delta)\}\right] \leq \sum_{n:\, h\in\boldsymbol{\mathcal{H}}_n} \mathbb{P}[L_{\boldsymbol{\mathcal{D}}}(h) > L_{\boldsymbol{\mathcal{S}}}(h) + \varepsilon_n(m, w(n)\delta)] \tag{1.16}$$

We can then apply Equation 1.12: the probability that the approximation error is higher than $\varepsilon_n(m, w(n)\delta)$ is, by definition, bounded by $w(n)\delta$. The bound then becomes

$$\mathbb{P}\left[\bigcup_{n:\, h\in\boldsymbol{\mathcal{H}}_n} \{L_{\boldsymbol{\mathcal{D}}}(h) > L_{\boldsymbol{\mathcal{S}}}(h) + \varepsilon_n(m, w(n)\delta)\}\right] \leq \sum_{n:\, h\in\boldsymbol{\mathcal{H}}_n} \delta\, w(n) \tag{1.17}$$

As δ is constant, we can move it outside of the sum:

$$\mathbb{P}\left[\bigcup_{n:\, h\in\boldsymbol{\mathcal{H}}_n}\{L_{\boldsymbol{\mathcal{D}}}(h) > L_{\boldsymbol{\mathcal{S}}}(h) + \varepsilon_n(m, w(n)\delta)\}\right] \leq \delta \sum_{n:\, h\in\boldsymbol{\mathcal{H}}_n} w(n) \tag{1.18}$$

Finally, we set condition $\sum_{n=1}^{\infty} w(n) \leq 1$, so we can bound this further as

$$\mathbb{P}\left[\bigcup_{n:\, h\in\boldsymbol{\mathcal{H}}_n}\{L_{\boldsymbol{\mathcal{D}}}(h) > L_{\boldsymbol{\mathcal{S}}}(h) + \varepsilon_n(m, w(n)\delta)\}\right] \leq \delta \tag{1.19}$$

This proves the second theorem statement in Equation 1.14. □

1.7 Perceptron Convergence

Theorem 1.7.1 (Perceptron Convergence Theorem)**.** *Assume that* $(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_m, y_m)$ *is separable, let* $B = \min\{\|\mathbf{w}\| : \forall i \in [m], y_i\langle \mathbf{w}, \mathbf{x}_i\rangle \geq 1\}$*, and let* $R = \max_i \|\mathbf{x_i}\|$*. Then, the Perceptron algorithm stops after at most* $(RB)^2$ *iterations, and when it stops it holds that* $\forall i \in [m]$*,* $y_i\langle \mathbf{w}^{(t)}, \mathbf{x}_i\rangle > 0$*.*

Proof. By the definition of the stopping condition, if the Perceptron stops it must have separated all the examples. We will show that if the Perceptron runs for T iterations, then we must have $T \leq (RB)^2$, which implies the Perceptron must stop after at most $(RB)^2$ iterations.

Let $\mathbf{w}^*$ be a vector that achieves the minimum in the definition of B. That is, $y_i\langle \mathbf{w}^*, \mathbf{x}_i\rangle \geq 1$ for all i, and among all vectors that satisfy these constraints, $\mathbf{w}^*$ is of minimal norm.

The idea of the proof is to show that after performing T iterations, the cosine of the angle between $\mathbf{w}^*$ and $\mathbf{w}^{(T+1)}$ is at least $\frac{\sqrt{T}}{RB}$. That is, we will show that

$$\frac{\langle \mathbf{w}^*, \mathbf{w}^{(T+1)}\rangle}{\|\mathbf{w}^*\|\|\mathbf{w}^{(T+1)}\|} \geq \frac{\sqrt{T}}{RB} \tag{1.20}$$

By the Cauchy-Schwartz inequality, the LHS of Equation 1.20 is at most 1. Therefore, Equation 1.20 would imply that

$$1 \geq \frac{\sqrt{T}}{RB} \quad \Rightarrow \quad T \leq (RB)^2$$

which will conclude our proof.

To show that Equation 1.20 holds, we first show that $\langle \mathbf{w}^*, \mathbf{w}^{(T+1)}\rangle \geq T$. Indeed, at the first iteration, $\mathbf{w}^{(1)} = (0, \ldots, 0)$ and therefore $\langle \mathbf{w}^*, \mathbf{w}^{(1)}\rangle = 0$, while on iteration t, if we update using example $(\mathbf{x}_i, y_i)$ we have that

$$\begin{aligned}\langle \mathbf{w}^*, \mathbf{w}^{(t+1)}\rangle - \langle \mathbf{w}^*, \mathbf{w}^{(t)}\rangle &= \langle \mathbf{w}^*, \mathbf{w}^{(t+1)} - \mathbf{w}^{(t)}\rangle = \\ &= \langle \mathbf{w}^*, y_i\mathbf{x}_i\rangle = y_i\langle \mathbf{w}^*, \mathbf{x}_i\rangle \geq 1\end{aligned}$$

Therefore, after performing T iterations, we get

$$\langle \mathbf{w}^*, \mathbf{w}^{(T+1)}\rangle = \sum_{t=1}^{T}\left(\langle \mathbf{w}^*, \mathbf{w}^{(t+1)}\rangle - \langle \mathbf{w}^*, \mathbf{w}^{(t)}\rangle\right) \geq T \tag{1.21}$$

as required.

Next, we upper bound $\|\mathbf{w}^{(T+1)}\|$. For each iteration t we have that

$$\begin{aligned}\|\mathbf{w}^{(t+1)}\|^2 &= \|\mathbf{w}^{(t)} + y_i\mathbf{x}_i\|^2 = \\ &= \|\mathbf{w}^{(t)}\|^2 + 2y_i\langle\mathbf{w}^{(t)}, \mathbf{x}_i\rangle + y_i^2\|\mathbf{x}_i\|^2 \leq \|\mathbf{w}^{(t)}\|^2 + R^2\end{aligned} \tag{1.22}$$

where the last inequality is due to the fact that example i is necessarily such that $y_i\langle\mathbf{w}^{(t)}, \mathbf{x}_i\rangle \leq 0$, and the norm of $\mathbf{x}_i$ is at most R. Now, since $\|\mathbf{w}^{(1)}\|^2 = 0$, if we use Equation 1.22 recursively for T iterations, we obtain that

$$\|\mathbf{w}^{(T+1)}\|^2 \leq TR^2 \quad \Rightarrow \quad \|\mathbf{w}^{(T+1)}\| \leq \sqrt{T}R \tag{1.23}$$

Combining Equation 1.21 with 1.23, and using the fact that $\|\mathbf{w}^*\| = B$, we obtain that

$$\frac{\langle\mathbf{w}^{(T+1)}, \mathbf{w}^*\rangle}{\|\mathbf{w}^*\|\|\mathbf{w}^{(T+1)}\|} \geq \frac{T}{B\sqrt{T}R} = \frac{\sqrt{T}}{BR}$$

We have thus shown that Equation 1.20 holds, and this concludes our proof.

□

1.8 VC-dimension of Halfspaces

Theorem 1.8.1 (VC-dimension of Halfspaces)**.** *The VC-dimension of the class of homogeneous halfspaces in $\mathbb{R}^d$ is d. The VC-dimension of the class of non-homogeneous halfspaces in $\mathbb{R}^d$ is $d+1$.*

Proof. (homogeneous case). To prove the homogeneous case, we show that

1. $\text{VCdim}(HS_d) \geq d$
2. $\text{VCdim}(HS_d) < d+1$

First, consider the set of vector $\mathbf{e}_1, \ldots, \mathbf{e}_d$ where for every i the vector $\mathbf{e}_i$ is the all zeros vector except 1 in the i'th coordinate. The set is shattered by the class of homogeneous halfspaces. Indeed, for every labeling $y_1, \ldots, y_d$, set $\mathbf{w} = (y_1, \ldots, y_d)$, and then $\langle\mathbf{w}, \mathbf{e}_i\rangle = y_i$ for all i.

Next, let $\mathbf{x}_1, \ldots \mathbf{x}_{d+1}$ be a set of $d+1$ vectors in $\mathbb{R}^d$. Then, there must exist real numbers $a_1, \ldots, a_{d+1}$, not all of them are zero, such that $\sum_{i=1}^{d+1} a_i\mathbf{x}_i = \mathbf{0}$. Let $I = \{i : a_i > 0\}$ and $J = \{j : a_j < 0\}$. Either I or J is non-empty. Let us first assume that both of them are non-empty. Then,

$$\sum_{i\in I} a_i\mathbf{x}_i = \sum_{j\in J} |a_j|\mathbf{x}_j$$

Now, suppose that $\mathbf{x}_1, \ldots, \mathbf{x}_{d+1}$ are shattered by the class of homogeneous classes. Then, there must exist a vector $\mathbf{w}$ such that $\langle\mathbf{w}, \mathbf{x}_i\rangle > 0$ for all $i \in I$ while $\langle\mathbf{w}, \mathbf{x}_j\rangle < 0$ for every $j \in J$. It follows that

$$0 < \sum_{i\in I} a_i\langle\mathbf{x}_i, \mathbf{w}\rangle = \left\langle\sum_{i\in I} a_i\mathbf{x}_i, \mathbf{w}\right\rangle = \left\langle\sum_{j\in J} |a_j|\mathbf{x}_j, \mathbf{w}\right\rangle = \sum_{j\in J} |a_j|\langle\mathbf{x}_j, \mathbf{w}\rangle < 0$$

which leads to a contradiction. Finally, if J (respectively, I) is empty, then the right-most (respectively, left-most) inequality should be replaced by an equality, which still leads to a contradiction. □

1.9 Optimality of the Least Squares Algorithm

Theorem 1.9.1 (Optimality of the Least Squares Algorithm)**.** *Consider a linear regression problem with domain set* $\boldsymbol{\mathcal{X}} \subset \mathbb{R}^d$ *and label set* $\boldsymbol{\mathcal{Y}} = \mathbb{R}$*. Let the hypothesis class be the set of linear functions*

$$\boldsymbol{\mathcal{H}}_{reg} = L_d = \{\mathbf{x} \mapsto \langle \mathbf{w}, \mathbf{x} \rangle + b : \mathbf{w} \in \mathbb{R}^d,\ b \in \mathbb{R}\}.$$

Let the loss function be the squared loss

$$\ell(h, (\mathbf{x}, y)) = (h(\mathbf{x}) - y)^2,$$

and let the empirical risk be

$$L_{\boldsymbol{\mathcal{S}}}(h) = \frac{1}{m} \sum_{i=1}^{m} (h(\mathbf{x}_i) - y_i)^2.$$

Then, the least squares algorithm returns a hypothesis

$$\hat{h} \in \arg\min_{h \in \boldsymbol{\mathcal{H}}_{reg}} L_{\boldsymbol{\mathcal{S}}}(h)$$

that is, it solves optimally the empirical risk minimization problem over $\boldsymbol{\mathcal{H}}_{reg}$ *with respect to the squared loss.*

Proof. The ERM approach to regression is solved by the least squares algorithm. Since the square is a quadratic function, we can treat it as a convex problem: it has a single minimum. Given a training set $\boldsymbol{\mathcal{S}}$ and using the homogeneous version of L_d, the ERM problem is to find

$$\arg\min_{\mathbf{w}} L_{\boldsymbol{\mathcal{S}}}(h_{\mathbf{w}}) = \arg\min_{\mathbf{w}} \frac{1}{m} \sum_{i=1}^{m} (\langle \mathbf{w}, \mathbf{x}_i \rangle - y_i)^2$$

Thus, we compute the gradient of the objective function and compare it to zero. That is, we need to solve

$$\frac{\partial L_{\boldsymbol{\mathcal{S}}}}{\partial \mathbf{w}} = \frac{2}{m} \sum_{i=1}^{m} (\langle \mathbf{w}, \mathbf{x}_i \rangle - y_i) \mathbf{x}_i = 0$$

We can rewrite the problem as $A\mathbf{w} = \mathbf{b}$, where

$$A = \sum_{i=1}^{m} x_i x_i^T, \qquad \mathbf{b} = \sum_{i=1}^{m} y_i \mathbf{x}_i$$

If A is invertible, then the solution to the ERM problem is

$$\mathbf{w} = A^{-1}\mathbf{b}$$

If A is not invertible, the system $A\mathbf{w} = \mathbf{b}$ admits infinitely many solutions. In this case, all minimizers achieve the same empirical risk. A canonical solution is obtained by selecting the minimizer of minimal norm, which is given by

$$\mathbf{w}^\star = X^+ \mathbf{y},$$

where X^+ denotes the Moore–Penrose pseudoinverse of the data matrix X.

□

1.10 Distance Theorem for Halfspaces

Theorem 1.10.1 (Distance Theorem for Halfspaces)**.** *The distance between a hyperplane P defined by $(\mathbf{w}, b)$, assuming $\|\mathbf{w}\| = 1$, and a point $\mathbf{x}$ is*

$$d(\mathbf{x}, P) = |\langle \mathbf{w}, \mathbf{x} \rangle + b| \tag{1.24}$$

Proof. The distance between a point $\mathbf{x}$ and the hyperplane is defined as

$$\min \{\|\mathbf{x} - \mathbf{v}\| \colon \langle \mathbf{w}, \mathbf{v} \rangle + b = 0\}$$

Taking $\mathbf{v} = \mathbf{x} - (\langle \mathbf{w}, \mathbf{x} \rangle + b)\mathbf{w}$, we can prove that it lies on the hyperplane. Thus, since $\|\mathbf{w}\|^2 = 1$, we have that

$$\langle \mathbf{w}, \mathbf{v} \rangle + b = \langle \mathbf{w}, \mathbf{x} \rangle - (\langle \mathbf{w}, \mathbf{x} \rangle + b)\|\mathbf{w}\|^2 + b = 0$$

and the distance between $\mathbf{x}$ and $\mathbf{v}$ is

$$\|\mathbf{x} - \mathbf{v}\| = |\langle \mathbf{w}, \mathbf{x} \rangle + b| \|\mathbf{w}\| = |\langle \mathbf{w}, \mathbf{x} \rangle + b|$$

Hence, the distance is at most $\|\mathbf{x} - \mathbf{v}\| = |\langle \mathbf{w}, \mathbf{x} \rangle + b|$. Since $\mathbf{v}$ is on the hyperplane, it is an upper bound to distance. We need to prove that it is the closest point. We then take a generic point $\mathbf{u}$ on the hyperplane, thus $\langle \mathbf{w}, \mathbf{u} \rangle + b = 0$. Expanding the square, we get

$$\begin{aligned}
\|\mathbf{x} - \mathbf{u}\|^2 &= \|\mathbf{x} - \mathbf{v} + \mathbf{v} - \mathbf{u}\|^2 = \\
&= \|\mathbf{x} - \mathbf{v}\|^2 + \|\mathbf{v} - \mathbf{u}\|^2 + 2\langle \mathbf{x} - \mathbf{v}, \mathbf{v} - \mathbf{u} \rangle \geq \\
&\geq \|\mathbf{x} - \mathbf{v}\|^2 + 2\langle \mathbf{x} - \mathbf{v}, \mathbf{v} - \mathbf{u} \rangle \\
&= \|\mathbf{x} - \mathbf{v}\|^2 + 2(\langle \mathbf{w}, \mathbf{x} \rangle + b)\langle \mathbf{w}, \mathbf{v} - \mathbf{u} \rangle = \|\mathbf{x} - \mathbf{v}\|^2
\end{aligned}$$

where the last equality is because $\langle \mathbf{w}, \mathbf{v} \rangle = \langle \mathbf{w}, \mathbf{u} \rangle = -b$. Hence, the distance between $\mathbf{x}$ and $\mathbf{u}$ is at least the distance between $\mathbf{x}$ and $\mathbf{v}$, which concludes our proof. □

1.11 Equivalence of the Convex SVM Problem

Theorem 1.11.1 (Equivalence of the Convex SVM Problem)**.** *A Hard-SVM tries to maximize the margin on linearly separable data. The optimization problem is*

$$\arg\max_{(\mathbf{w}, b) \colon \|\mathbf{w}\| = 1} \min_{i \in [m]} y_i(\langle \mathbf{w}, \mathbf{x}_i \rangle + b). \tag{1.25}$$

We can find an equivalent formulation, which is a convex problem:

$$\arg\min_{\mathbf{w}, b} \|\mathbf{w}\|^2 \quad s.t. \quad y_i(\langle \mathbf{w}, \mathbf{x}_i \rangle + b) \geq 1, \quad \forall i \in [m]. \tag{1.26}$$

This problem can be solved using standard tools, e.g. Lagrange multipliers or gradient descent.

Proof. We start from the hard-margin SVM formulation:

$$\arg\max_{(\mathbf{w}, b) \colon \|\mathbf{w}\| = 1} \min_{(x_i, y_i) \in \boldsymbol{\mathcal{S}}} y_i(\langle \mathbf{w}, \mathbf{x}_i \rangle + b).$$

Let $(\mathbf{w}^*, b^*)$ be an optimal solution of this problem and define the optimal margin

$$\gamma^* = \min_{(x_i,y_i)\in\boldsymbol{\mathcal{S}}} y_i(\langle \mathbf{w}^*, \mathbf{x}_i\rangle + b^*).$$

By definition, for all $(x_i, y_i) \in \boldsymbol{\mathcal{S}}$ we have

$$y_i(\langle \mathbf{w}^*, \mathbf{x}_i\rangle + b^*) \geq \gamma^*.$$

Dividing both sides by γ^* yields

$$y_i\left(\left\langle \frac{\mathbf{w}^*}{\gamma^*}, \mathbf{x}_i \right\rangle + \frac{b^*}{\gamma^*}\right) \geq 1,$$

which shows that the scaled pair $(\mathbf{w}^*/\gamma^*,\ b^*/\gamma^*)$ is a feasible solution of the convex optimization problem

$$\min_{\mathbf{w},b} \|\mathbf{w}\|^2 \quad \text{s.t.} \quad y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b) \geq 1 \quad \forall i.$$

Now let $(\mathbf{w}_0, b_0)$ be an optimal solution of the convex problem above. By optimality, we must have

$$\|\mathbf{w}_0\| \leq \left\| \frac{\mathbf{w}^*}{\gamma^*} \right\| = \frac{1}{\gamma^*},$$

since otherwise $(\mathbf{w}^*/\gamma^*, b^*/\gamma^*)$ would yield a smaller objective value.

The geometric margin achieved by $(\mathbf{w}_0, b_0)$ is

$$\min_i \frac{y_i(\langle \mathbf{w}_0, \mathbf{x}_i\rangle + b_0)}{\|\mathbf{w}_0\|} \geq \frac{1}{\|\mathbf{w}_0\|} \geq \gamma^*.$$

Hence, the solution of the convex problem attains a margin at least as large as the optimal hard-margin SVM margin. Since γ^* is maximal by definition, the two optimization problems are equivalent and yield the same optimal separating hyperplane. □

1.12 Representer Theorem

The soft SVM problem in feature space can be rewritten as

$$\min_{\mathbf{w}}(f(\langle \mathbf{w}, \psi(\mathbf{x}_1)\rangle, \ldots, \langle \mathbf{w}, \psi(\mathbf{x}_m)\rangle) + R(\|\mathbf{w}\|))$$

where $R(\|\mathbf{w}\|)$ is a monotonically non-decreasing function.

For Soft-SVM $R(a) = \lambda a^2$ and $f(a_1, \ldots, a_m) = \frac{1}{m}\sum_i \max\{0, 1 - y_i a_i\}$. For Hard-SVM for non-homogeneous halfspaces $R(a) = a^2$ and

$$f(a_1, \ldots, a_m) = \begin{cases} 0 & \text{if } \exists b : y_i(a_i + b) \geq 1\, \forall i \\ \infty & \text{otherwise} \end{cases}$$

Theorem 1.12.1 (Representer Theorem)**.** *Assume that ψ is a mapping from $\boldsymbol{\mathcal{X}}$ to a Hilbert space. Then, there exists a vector $\boldsymbol{\alpha} \in \mathbb{R}^m$ such that*

$$\mathbf{w}^* = \sum_{i=1}^{m} \alpha_i \psi(\mathbf{x}_i) \tag{1.27}$$

is an optimal solution of the optimization problem

$$\min_{\mathbf{w}}(f(\langle \mathbf{w}, \psi(\mathbf{x}_1)\rangle, \ldots, \langle \mathbf{w}, \psi(\mathbf{x}_m)\rangle) + R(\|\mathbf{w}\|)) \tag{1.28}$$

In other words, an optimal solution of Equation 1.28 is a linear combination of vectors in the feature space with a set of coefficients α_i.

Proof. Let $\mathbf{w}^*$ be an optimal solution of Equation 1.28. Since $\mathbf{w}^*$ belongs to a Hilbert space, we can rewrite $\mathbf{w}^*$ as

$$\mathbf{w}^* = \sum_{i=1}^{m} \alpha_i \psi(\mathbf{x}_i) + \mathbf{u}$$

where $\langle \mathbf{u}, \psi(\mathbf{x}_i) \rangle = 0$ for all i. Set $\mathbf{w} = \mathbf{w}^* - \mathbf{u}$. Clearly, $\|\mathbf{w}^*\|^2 = \|\mathbf{w}\|^2 + \|\mathbf{u}\|^2$, thus $\|\mathbf{w}\| \leq \|\mathbf{w}^*\|$. Since R is non-decreasing, we obtain $R(\|\mathbf{w}\|) \leq R(\|\mathbf{w}^*\|)$. Additionally, for all i we have that

$$y_i \langle \mathbf{w}, \psi(\mathbf{x}_i) \rangle = y_i \langle \mathbf{w}^* - \mathbf{u}, \psi(\mathbf{x}_i) \rangle = y_i \langle \mathbf{w}^*, \psi(\mathbf{x}_i) \rangle$$

hence

$$f(y_1 \langle \mathbf{w}, \psi(\mathbf{x}_1) \rangle, \ldots, y_m \langle \mathbf{w}, \psi(\mathbf{x}_m) \rangle) = f(y_1 \langle \mathbf{w}^*, \psi(\mathbf{x}_1) \rangle, \ldots, y_m \langle \mathbf{w}^*, \psi(\mathbf{x}_m) \rangle)$$

We have shown that the objective of Equation 1.28 at $\mathbf{w}$ cannot be larger than the objective at $\mathbf{w}^*$ and therefore $\mathbf{w}$ is an optimal solution. Since $\mathbf{w} = \sum_{i=1}^{m} \alpha_i \psi(\mathbf{x}_i)$ □

1.13 Polynomial Kernel Derivation

The N degree polynomial kernel is defined to be

$$K(\mathbf{x}, \mathbf{x}') = (1 + \langle \mathbf{x}, \mathbf{x}' \rangle)^N \tag{1.29}$$

Proof. We will prove that this is indeed a kernel function. We will show that there exists a mapping ψ from the original space to some higher dimensional space for which $K(\mathbf{x}, \mathbf{x}') = \langle \psi(\mathbf{x}), \psi(\mathbf{x}') \rangle$.

Denoting $x_0 = x'_0 = 1$ (homogeneous trick), we can expand the kernel

$$(1 + \langle \mathbf{x}, \mathbf{x}' \rangle)^N = \left(\sum_{i=0}^{d} x_i x'_i \right)^N = \prod_{n=1}^{N} \sum_{i=0}^{d} x_i x'_i =$$
$$= \prod_{n=1}^{N} (1 + x_1 x'_1 + x_2 x'_2 + \ldots + x_d x'_d)$$

The product of polynomials is given by the combination of sums (remember that $J = \{0, 1, \ldots, d\}^N$)

$$(1 + \langle \mathbf{x}, \mathbf{x}' \rangle)^N = \sum_{j \in J} \prod_{i=1}^{N} x_{j_i} x'_{j_i}$$

Each element of the feature space embedding is given by $\psi_j = \prod_{i=1}^{N} x_{j_i}$. Therefore, we can trivially see that $\langle \psi(\mathbf{x}), \psi(\mathbf{x}') \rangle = K(\mathbf{x}, \mathbf{x}')$. □

1.14 RBF Kernel Derivation

The Radial Basis Function (RBF) kernel has a Gaussian form as a function of the distance. The Gaussian kernel is defined to be

$$K(\mathbf{x}, \mathbf{x}') = \exp\left\{ -\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2\sigma^2} \right\} \tag{1.30}$$

We can control the precision (how close it is to the points) by changing the variance.

Proof. For a 1D input, we define a function

$$\psi_n(x) = \frac{1}{\sqrt{n!}} e^{-\frac{x^2}{2}} x^n$$

where we considered the RB kernel with $\sigma = 1$.

The feature space is infinite-dimensional for $n \in \{0, 1, \ldots, \infty\}$. We have

$$\begin{aligned}\langle \psi(x), \psi(x') \rangle &= \sum_{n=0}^{\infty} \left(\frac{1}{\sqrt{n!}} e^{-\frac{x^2}{2}} x^n \right) \left(\frac{1}{\sqrt{n!}} e^{-\frac{x'^2}{2}} x'^n \right) = \\ &= e^{-\frac{x^2 + x'^2}{2}} \sum_{n=0}^{\infty} \frac{(xx')^n}{n!}\end{aligned}$$

The infinite series converges to the exponential function $\sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x$. Therefore,

$$\langle \psi(x), \psi(x') \rangle = e^{-\frac{x^2 + x'^2 - 2xx'}{2}} = e^{-\frac{\|x - x'\|^2}{2}}$$

which is equivalent to the RBF kernel function. □

1.15 Weak Learnability Property of Decision Stumps

Definition 1.15.1 (Weak Learnability)**.** A hypothesis class $\boldsymbol{\mathcal{H}}$ is γ**-weak** learnable if there is a function $m_{\boldsymbol{\mathcal{H}}} : (0, 1)^2 \to \mathbb{N}$ and a learning algorithm such that for every $\delta \in (0, 1)$ and for every learning problem $\boldsymbol{\mathcal{D}}$, f with realizability assumption, the algorithm will satisfy the following condition over an iid training set of size $m \geq m_{\boldsymbol{\mathcal{H}}}$:

$$L_{\boldsymbol{\mathcal{D}},f}(h) \leq \frac{1}{2} - \gamma \tag{1.31}$$

Theorem 1.15.2. *Decision stumps are weak learners.*

Proof. Consider binary classification on $\mathbb{R}$ and the class of decision stumps

$$B = \{x \mapsto b\,\mathrm{sign}(x - \theta) : b \in \{-1, 1\},\ \theta \in \mathbb{R}\}.$$

Every classifier obtained by combining decision stumps partitions the real line into three consecutive regions. For any pair of regions, there exists a decision stump that agrees exactly with the labels on those two regions.

Since there are three regions, at least one pair of regions must contain at least 2/3 of the data points. Therefore, there exists a decision stump whose classification error is at most 1/3.

The VCdim of the stump is 2, so it is learnable by agnostic PAC.

□

1.16 VC-dimension of Boosted Solutions

Lemma 1.16.1 (Sauer's Lemma)**.** *Let* $\boldsymbol{\mathcal{H}}$ *be a class with* $\mathrm{VCdim}\,(\boldsymbol{\mathcal{H}}) \leq d < \infty$*. Then, for all* m*,* $\tau_{\boldsymbol{\mathcal{H}}}(m) \leq \sum_{i=0}^{d} \binom{m}{i}$*. In particular, if* $m > d + 1$*, then*

$$\tau_{\boldsymbol{\mathcal{H}}}(m) \leq \left(\frac{em}{d} \right)^d \tag{1.32}$$

Theorem 1.16.2 (VC-dimension of Boosted Solutions)**.** *If B is a base class and $L(B,T)$ is a boosted solution, with* $\text{VCdim}(B) \geq 3$ *and $T \geq 3$, we have*

$$\text{VCdim}(L(B,T)) \leq T(\text{VCdim}(B)+1)(3\log(T(\text{VCdim}(B)+2))+2) \tag{1.33}$$

Proof. Let $C = \{x_1, \ldots, x_m\}$ be a set shattered by $L(B,T)$.

1. We choose $h_1, \ldots, h_T \in B$
2. We apply a halfspace hypothesis $\mathbf{w}$

By Sauer's lemma ((1.32)), the number of distinct labelings of C induced by the base class B is at most $\left(\frac{em}{d}\right)^d$. Therefore, the number of possible labelings induced by choosing T hypotheses $h_1, \ldots, h_T \in B$ is at most $\left(\frac{em}{d}\right)^{dT}$.

When we apply the linear predictor in $\mathbb{R}^T$, we use the fact that the VC-dimension of the class of halfspaces is $T+1$. By Sauer's lemma, there are at most $\left(\frac{em}{T}\right)^T$ different dichotomies (the VC-dimension of the halfspace hypothesis class is equal to the dimension in homogeneous coordinates).

The overall number of dichotomies is bounded by

$$\tau_{L(B,T)}(m) \leq \left(\frac{em}{d}\right)^{dT}\left(\frac{em}{T}\right)^T$$

Since $T \geq 3$ and $d \geq 3$, we have $\frac{e}{T} < 1$ and $\frac{e}{d} < 1$. Therefore,

$$\left(\frac{em}{d}\right)^{dT}\left(\frac{em}{T}\right)^T < m^{(d+1)T}$$

We have fewer than $m^{(d+1)T}$ different functions, but we need to have 2^m to shatter the set:

$$m^{(d+1)T} \geq 2^m$$

We can turn it around to get

$$m \leq \log(m)\frac{(d+1)T}{\log(2)}$$

We know that $x < a\log(x) \quad \Rightarrow \quad x < 2a\log(a)$, therefore

$$m \leq \frac{2(d+1)T}{\log(2)}\log\left(\frac{(d+1)T}{\log(2)}\right)$$

Moreover, we know that $\log(2) \simeq 0.69$. Thus, we can just substitute 2 and 3 for the parts to get a looser bound with whole numbers

$$m \leq (d+1)T(3\log((d+1)T)+2)$$

This is the theorem statement. □

1.17 SGD Bound for RLM

Definition 1.17.1 (Lipschitzness)**.** Let $C \subset \mathbb{R}^d$. A function $f : \mathbb{R}^d \to \mathbb{R}^k$ is ρ-Lipschitz over C if for every $\mathbf{w}_1, \mathbf{w}_2 \in C$ we have that

$$\|f(\mathbf{w}_1) - f(\mathbf{w}_2)\| \leq \rho\|\mathbf{w}_1 - \mathbf{w}_2\|. \tag{1.34}$$

Definition 1.17.2 (Strongly Convex Function)**.** A function f is λ-strongly convex if for all $\mathbf{w}$, $\mathbf{u}$ and $\alpha \in (0,1)$ we have

$$f(\alpha\mathbf{w} + (1-\alpha)\mathbf{u}) \leq \alpha f(\mathbf{w}) + (1-\alpha)f(\mathbf{u}) - \frac{\lambda}{2}\alpha(1-\alpha)\|\mathbf{w}-\mathbf{u}\|^2 \tag{1.35}$$

Theorem 1.17.3. *Assume that f is λ-strongly convex and that $\mathbb{E}[\|\mathbf{v}_t\|^2] \leq \rho^2$. Let $\mathbf{w}^* \in \operatorname{argmin}_{\mathbf{w}\in\boldsymbol{\mathcal{H}}} f(\mathbf{w})$ be an optimal solution. Then,*

$$\mathbb{E}[f(\bar{\mathbf{w}})] - f(\mathbf{w}^*) \leq \frac{\rho^2}{2\lambda T}(1+\log(T)) \tag{1.36}$$

Theorem 1.17.4 (SGD bound for RLM)**.** *The optimization problem associated with RLM is*

$$\min_{\mathbf{w}} \frac{\lambda}{2}\|\mathbf{w}\|^2 + L_{\boldsymbol{\mathcal{S}}}(\mathbf{w}) \tag{1.37}$$

If the loss is ρ-Lipschitz, after T iterations we get:

$$\mathbb{E}[f(\bar{\mathbf{w}})] - f(\mathbf{w}^*) \leq \frac{4\rho^2}{\lambda T}(1+\log(T)) \tag{1.38}$$

Proof. Define

$$f(\mathbf{w}) = \frac{\lambda}{2}\|\mathbf{w}\|^2 + L_{\boldsymbol{\mathcal{S}}}(\mathbf{w}) \tag{1.39}$$

which is λ-strongly convex function; therefore, we can apply SGD variant for minimizing a λ-strongly convex function with $\boldsymbol{\mathcal{H}} = \mathbb{R}^d$. To apply it, we only need to find a way to construct an unbiased estimate of a subgradient of f at $\mathbf{w}^{(t)}$. This is easily done by noting that if we pick z uniformly at random from $\boldsymbol{\mathcal{S}}$, and choose $\mathbf{v}_t \in \partial\ell(\mathbf{w}^{(t)}, z)$ then the expected value of $\lambda\mathbf{w}^{(t)} + \mathbf{v}_t$ is a subgradient of f at $\mathbf{w}^{(t)}$.

To analyze the resulting algorithm, we first rewrite the update rule

$$\begin{aligned}
\mathbf{w}^{(t+1)} &= \mathbf{w}^{(t)} - \frac{\lambda\mathbf{w}^{(t)} + \mathbf{v}t}{\lambda t} = \\
&= \frac{t-1}{t}\mathbf{w}^{(t)} - \frac{\mathbf{v}_t}{\lambda t} = \\
&= \frac{t-1}{t}\left(\frac{t-2}{t-1}\mathbf{w}^{(t-1)} - \frac{\mathbf{v}_{t-1}}{\lambda(t-1)}\right) - \frac{\mathbf{v}_t}{\lambda t} \\
&= -\frac{1}{\lambda t}\sum_{k=1}^{t}\mathbf{v}_k
\end{aligned}$$

where in the last step we applied the rule recursively.

If we assume that the loss function is ρ-Lipschitz, it follows that for all t we have $\|\mathbf{v}_t\| \leq \rho$ and therefore $\|\lambda\mathbf{w}_t\| \leq \rho$, which yields

$$\|\lambda\mathbf{w}^{(t)} + \mathbf{v}_t\| \leq 2\rho$$

Therefore, Theorem 1.17.3 tells us that after performing T iterations, we have that

$$\mathbb{E}[f(\bar{\mathbf{w}})] - f(\mathbf{w}^*) \leq \frac{4\rho^2}{\lambda T}(1+\log(T))$$

1.18 Derivation of Backpropagation

Our goal in ERM is to minimize the empirical risk

$$\mathbf{w}^* = \arg\min_{\mathbf{w}} \frac{1}{m} \sum_{i=1}^{m} \ell(y_i, h_{\mathbf{w}}(\mathbf{x}_i)). \tag{1.40}$$

To optimize the weights using SGD, we need the gradient of the loss with respect to all the weights in the network. However, the output of the network is computed only at the last layer, while the weights are distributed across all layers. Backpropagation provides an efficient way to compute these gradients by repeatedly applying the chain rule.

Forward pass

For each layer t we define

$$a_j^{(t)} = \sum_i w_{j,i}^{(t)} o_i^{(t-1)}, \qquad o_j^{(t)} = f\left(a_j^{(t)}\right), \tag{1.41}$$

where $a_j^{(t)}$ is the pre-activation (linear combination) and $o_j^{(t)}$ is the output of neuron j at layer t.

Let the loss be $L(o^{(T)}, y)$.

Gradient with respect to the weights

We want to compute

$$\frac{\partial L}{\partial w_{j,i}^{(t)}}.$$

Since $w_{j,i}^{(t)}$ affects the loss through $a_j^{(t)}$, by the chain rule we obtain

$$\frac{\partial L}{\partial w_{j,i}^{(t)}} = \frac{\partial L}{\partial a_j^{(t)}} \frac{\partial a_j^{(t)}}{\partial w_{j,i}^{(t)}}. \tag{1.42}$$

Because

$$\frac{\partial a_j^{(t)}}{\partial w_{j,i}^{(t)}} = o_i^{(t-1)},$$

we define the *error signal* (delta)

$$\delta_j^{(t)} \triangleq \frac{\partial L}{\partial a_j^{(t)}}, \tag{1.43}$$

and obtain the compact expression

$$\frac{\partial L}{\partial w_{j,i}^{(t)}} = \delta_j^{(t)} \, o_i^{(t-1)}. \tag{1.44}$$

Therefore, once the deltas are known, all gradients can be computed using the activations from the forward pass.

Delta at the output layer

Let the final output be

$$y_m = g(a_m^{(T)}).$$

Then

$$\delta_m^{(T)} = \frac{\partial L}{\partial a_m^{(T)}} = \frac{\partial L}{\partial y_m} g'\left(a_m^{(T)}\right). \tag{1.45}$$

This gives the base case of the recursion.

Delta for a generic layer

The activation $a_j^{(t)}$ influences all neurons k in layer $t+1$. Applying the chain rule,

$$\delta_j^{(t)} = \frac{\partial L}{\partial a_j^{(t)}} = \sum_k \frac{\partial L}{\partial a_k^{(t+1)}} \frac{\partial a_k^{(t+1)}}{\partial a_j^{(t)}}. \tag{1.46}$$

Since

$$a_k^{(t+1)} = \sum_m w_{k,m}^{(t+1)} o_m^{(t)} \quad \text{and} \quad o_j^{(t)} = f(a_j^{(t)}),$$

we have

$$\frac{\partial a_k^{(t+1)}}{\partial a_j^{(t)}} = w_{k,j}^{(t+1)} f'\left(a_j^{(t)}\right). \tag{1.47}$$

Substituting into the previous expression yields the recursive rule

$$\delta_j^{(t)} = f'\left(a_j^{(t)}\right) \sum_k w_{k,j}^{(t+1)} \delta_k^{(t+1)}. \tag{1.48}$$

Backpropagation algorithm

The gradients are computed recursively as follows:

- **Base step:**

$$\delta^{(T)} = \frac{\partial L}{\partial y} \odot g'(a^{(T)}).$$

- **Recursive step:**

$$\delta^{(t)} = f'(a^{(t)}) \odot \left(W^{(t+1)\top} \delta^{(t+1)}\right).$$

- **Gradient:**

$$\frac{\partial L}{\partial W^{(t)}} = \delta^{(t)} (o^{(t-1)})^\top.$$

Thus, the error is propagated backward from the output layer to the input layer, and the gradients with respect to all weights can be computed efficiently in time proportional to the size of the network.

1.19 Optimal Inverse Lemma for PCA

Theorem 1.19.1 (Optimal Inverse Lemma for PCA)**.** *The PCA problem is*

$$\arg\min_{W\in\mathbb{R}^{n\times d},\, U\in\mathbb{R}^{d\times n}} \sum_{i=1}^{m} \|\mathbf{x}_i - UW\mathbf{x}_i\|^2 \tag{1.49}$$

Let (U, W) be the optimal solution of Equation 1.49. Then, the columns of U are orthonormal (namely, U^TU is the identity matrix in $\mathbb{R}^n$) and $W = U^T$. This means that we only need to find U, and then we can get both matrices almost for free.

Proof. Fix any U, W and consider the mapping $\mathbf{x} \mapsto UW\mathbf{x}$. The range of this mapping, $R = \{UW\mathbf{x} : \mathbf{x} \in \mathbb{R}^d\}$, is an n-dimensional linear subspace of $\mathbb{R}^d$. Let $V \in \mathbb{R}^{d\times n}$ be a matrix whose columns form an orthonormal basis of this subspace. Namely, the range of V is R, and $V^TV = I$. Therefore, each vector in R can be written as $V\mathbf{y}$ where $\mathbf{y} \in \mathbb{R}^n$. For every $\mathbf{x} \in \mathbb{R}^d$ and $\mathbf{y} \in \mathbb{R}^n$, we have

$$\|\mathbf{x} - V\mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \mathbf{y}^TV^TV\mathbf{y} - 2\mathbf{y}^TV^T\mathbf{x} = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\mathbf{y}^T(V^T\mathbf{x})$$

where we used the fact that V^TV is the identity matrix of $\mathbb{R}^n$. Minimizing the preceding expression with respect to $\mathbf{y}$ by comparing the gradient with respect to $\mathbf{y}$ to zero gives that $\mathbf{y} = V^T\mathbf{x}$. Therefore, for each $\mathbf{x}$ we have that

$$VV^T\mathbf{x} = \arg\min_{\tilde{\mathbf{x}}\in R} \|\mathbf{x} - \tilde{\mathbf{x}}\|^2$$

In particular, this holds for $\mathbf{x}_1, \ldots, \mathbf{x}_m$ and therefore we can replace (U, W) by (V, V^T) and by that we do not increase the objective

$$\sum_{i=1}^{m} \|\mathbf{x}_i - UW\mathbf{x}_i\|^2 \geq \sum_{i=1}^{m} \|\mathbf{x}_i - VV^T\mathbf{x}_i\|^2$$

and the proof follows. □

1.20 Correctness of K-means Clustering

Lemma 1.20.1 (Optimal Centroid)**.** *For a given cluster C_k, the centroid μ_k^* that minimizes the total distance is*

$$\mu_k^* = \frac{1}{|C_k|} \sum_{\mathbf{x}\in C_k} \mathbf{x} \tag{1.50}$$

The best centroid is just the barycenter of the points in the cluster.

Proof. The cost over the cluster is defined as

$$G(C_k, \mu_k) = \sum_{\mathbf{x}\in C_k} \|\mathbf{x} - \mu_k\|^2$$

The gradient over the centroid is

$$\frac{\partial G(C_k, \mu_k)}{\partial \mu_k} = \sum_{\mathbf{x}\in C_k} 2(\mu_k - \mathbf{x})$$

Setting the gradient to zero yields

$$\sum_{\mathbf{x}\in C_k} 2(\mu_k - \mathbf{x}) = 0 \quad \Rightarrow \quad |C_k|\mu_k = \sum_{\mathbf{x}\in C_k} \mathbf{x}$$

□

Theorem 1.20.2 (K-means Iterations)**.** *For each iteration of the K-means algorithm, the overall cost does not increase.*

Proof. We have already proved Theorem 1.20.1: the centroid computation minimizes the distance with respect to a given cluster allocation.

Since the selection of clusters uses argmin, it does not increase the total cost by definition.

Formally, the total cost is defined as

$$G(\mathcal{C}, \mu) = \sum_{k=1}^{K} \sum_{\mathbf{x}\in C_k} \|\mathbf{x} - \boldsymbol{\mu}_k\|^2$$

Cluster selection: we assign each point to the closest cluster (but keep the old centroids)

$$G(\mathcal{C}^{(t)}, \mu^{(t-1)}) = \sum_{k=1}^{K} \sum_{\mathbf{x}\in C_k^{(t)}} \|\mathbf{x} - \boldsymbol{\mu}^{(t-1)}\|^2 \leq \sum_{k=1}^{K} \sum_{\mathbf{x}\in C_k^{(t-1)}} \|\mathbf{x} - \boldsymbol{\mu}^{(t-1)}\|^2$$

Since each point is assigned to the closest centroid, the contribution of every point to the cost function cannot increase. Thus, we showed that

$$G(\mathcal{C}^{(t)}, \mu^{(t-1)}) \leq G(\mathcal{C}^{(t-1)}, \mu^{(t-1)})$$

Now we use Theorem 1.20.1: the selection of the new centroids is the one that minimizes the total distance for each cluster

$$\sum_{k=1}^{K} \sum_{\mathbf{x}\in C_k^{(t)}} \|\mathbf{x} - \boldsymbol{\mu}^{(t)}\|^2 \leq \sum_{k=1}^{K} \sum_{\mathbf{x}\in C_k^{(t)}} \|\mathbf{x} - \boldsymbol{\mu}^{(t-1)}\|^2$$

Putting it all together, each step reduces (or at least maintains) the cost function

$$G(\mathcal{C}^{(t)}, \mu^{(t)}) \leq G(\mathcal{C}^{(t)}, \mu^{(t-1)}) \leq G(\mathcal{C}^{(t-1)}, \mu^{(t-1)})$$

Therefore, each iteration of the K-means algorithm does not increase the value of the objective function. □